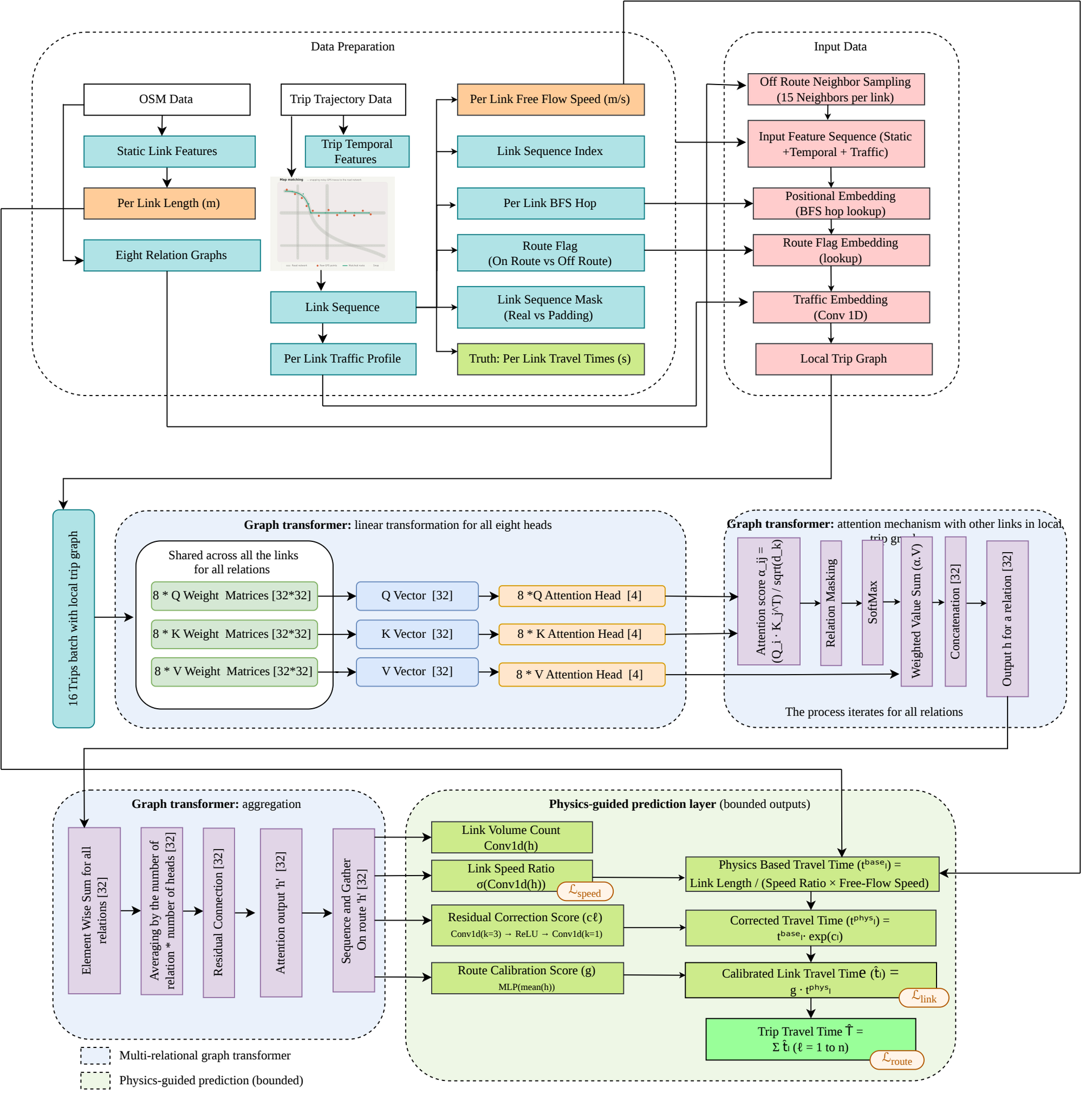

Data Preparation
OSM Data
Static Link Features
Per Link Length (m)
Eight Relation Graphs
Trip Trajectory Data
Trip Temporal Features
Map matching
Link Sequence
Per Link Traffic Profile
Per Link Free Flow Speed (m/s)
Link Sequence Index
Per Link BFS Hop
Route Flag
(On Route vs Off Route)
Link Sequence Mask
(Real vs Padding)
Truth: Per Link Travel Times (s)
Input Data
Off Route Neighbor Sampling
(15 Neighbors per link)
Input Feature Sequence (Static
+Temporal + Traffic)
Positional Embedding
(BFS hop lookup)
Route Flag Embedding
(lookup)
Traffic Embedding
(Conv 1D)
Local Trip Graph
16 Trips batch with local trip graph
Graph transformer: linear transformation for all eight heads
Shared across all the links
for all relations
8 * Q Weight Matrices [32*32]
8 * K Weight Matrices [32*32]
8 * V Weight Matrices [32*32]
Q Vector [32]
K Vector [32]
V Vector [32]
8 *Q Attention Head [4]
8 * K Attention Head [4]
8 * V Attention Head [4]
Graph transformer: attention mechanism with other links in local trip gr
Attention score α_ij =
(Q_i · K_j^T) / sqrt(d_k)
Relation Masking
SoftMax
Weighted Value Sum (α.V)
Concatenation [32]
Output h for a relation [32]
The process iterates for all relations
Graph transformer: aggregation
Element Wise Sum for all relations [32]
Averaging by the number of relation * number of heads [32]
Residual Connection [32]
Attention output 'h' [32]
Sequence and Gather On route 'h' [32]
Physics-guided prediction layer (bounded outputs)
Link Volume Count
Conv1d(h)
Link Speed Ratio
σ(Conv1d(h))
$\mathcal{L}_{speed}$
Residual Correction Score (cℓ)
Conv1d(k=3) → ReLU → Conv1d(k=1)
Route Calibration Score (g)
MLP(mean(h))
Physics Based Travel Time (t^{base}_l) =
Link Length / (Speed Ratio × Free-Flow Speed)
Corrected Travel Time (t^{phys}_l) =
$t^{base}_l \cdot \exp(c_l)$
Calibrated Link Travel Time ($\hat{t}_l$) =
$g \cdot t^{phys}_l$
$\mathcal{L}_{link}$
Trip Travel Time $\hat{T}$ =
$\Sigma\, \hat{t}_l$ (ℓ = 1 to n)
$\mathcal{L}_{route}$
Multi-relational graph transformer
Physics-guided prediction (bounded)